

Jerry Liu

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Interests: Long-context architectures; memory & knowledge storage in language models; in-context learning & algorithmic reasoning; ML for numerical & scientific computation.

EDUCATION

Stanford University — Ph.D. Candidate, Computational & Mathematical Engineering Sep 2022 — Spring 2027 (expected)
Advisor: Chris Ré @ Hazy Lab

- DOE Computational Science Graduate Fellowship.

Duke University — B.S. Mathematics & Computer Science, Minor in Music Aug 2018 — Dec 2021

RESEARCH & INDUSTRY EXPERIENCE

Stanford University, Hazy Research — PhD Researcher Sep 2022 — Present
Advisor: Chris Ré Stanford, CA

- Developed a Transformer-compatible fact-storing MLP with optimal storage scaling, using $10\text{--}104\times$ fewer parameters than prior constructions at matched fact counts (COLM 2026).
- Designing and evaluating long-context architectures that combine linear attention with explicit memory mechanisms for efficient retrieval and reasoning over long sequences.
- Designed BWLer for PINNs, improving benchmark accuracy by up to three orders of magnitude and earning a Best Paper Award at TASC @ COLT 2025.

Lawrence Berkeley National Lab, Machine Learning & Analytics — Research Intern Jun 2023 — Sep 2023
Advisors: N. Benjamin Erichson, Michael W. Mahoney Berkeley, CA

- Analyzed transformer architectures for **algorithmic reasoning in differential equation systems**, focusing on in-context system identification.
- Established theoretical and empirical connections between **attention mechanisms and pseudospectral PDE solvers** and evaluated generalization under distribution shifts

Duolingo, Learning AI Lab — Software Engineer Jan 2022 — Sep 2022
Pittsburgh, PA

- Enhanced user knowledge ML model serving **9 million DAUs** with multilingual LLMs
- Implemented tools for model performance visualization; productionized data cleaning and model training pipelines

Lawrence Livermore National Lab — Research Intern Jun 2021 — Dec 2021
Advisors: Jin Yao, Kenneth Weiss Livermore, CA

- Developed interface reconstruction algorithms with geometric optimization schemes to solve incompressible Navier-Stokes equations
- Formulated and implemented numerical schemes for advection with hybrid reconstruction-tracking method for multi-material fluid flow problems

SELECTED PUBLICATIONS

- Roberto Garcia*, **Jerry Liu***, Ronny Junkins*, Sabri Eyuboglu, Atri Rudra, Chris Ré. *MLPs are Hebbians: Constructing Efficient Fact-Storing MLPs for Transformers*. COLM 2026. [[paper](#) | [code](#) | [blog](#)]
- Catherine Deng*, Junmiao Hu*, Milan Rohatgi, **Jerry Liu**, Chris Ré. *Constructing Machine-Precision Neural Networks with Quasi-Interpolants*. AI&PDE @ ICLR 2026 **Oral (8/113 accepted papers)**. [[paper](#)]
- **Jerry Liu**, Yasa Baig, Denise Hui Jean Lee, Rajat Vadiraj Dwaraknath, Atri Rudra, Chris Ré. *BWLer: Barycentric Weight Layer Elucidates a Precision-Conditioning Tradeoff for PINNs*. TASC @ COLT 2025 (**Best Paper Award**) [[paper](#) | [code](#) | [blog](#)]
- **Jerry Liu**, Jessica Grogan, Owen Dugan, Ashish Rao, Simran Arora, Atri Rudra, Chris Ré. *Towards Learning High-Precision Least Squares Algorithms with Sequence Models*. ICLR 2025. [[paper](#) | [code](#)]
- **Jerry Liu**, N. Benjamin Erichson, Kush Bhatia, Michael W. Mahoney, Chris Ré. *Does In-Context Operator Learning Generalize to Domain-Shifted Settings?* DLDE III, NeurIPS 2023. [[paper](#)]

SKILLS

Machine Learning PyTorch, JAX, Hugging Face; model training and evaluation; sequence modeling, in-context learning, long-context and memory architectures, scientific ML

Tools NumPy, SciPy, Pandas, Git, Slurm, Weights & Biases

Mathematics Numerical linear algebra, optimization, numerical PDEs, statistical learning theory

Programming Python, C++, Java, MATLAB, SQL